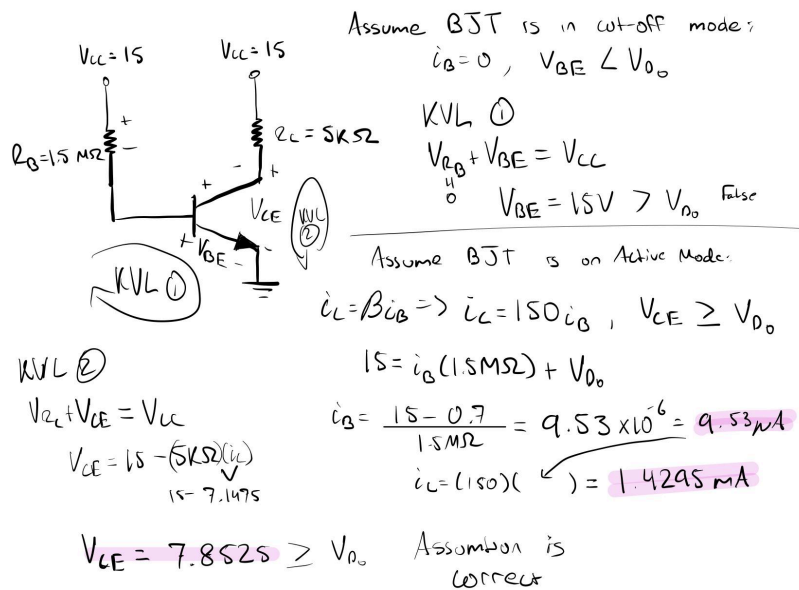


Pre-Lab 6- BJT Biasing

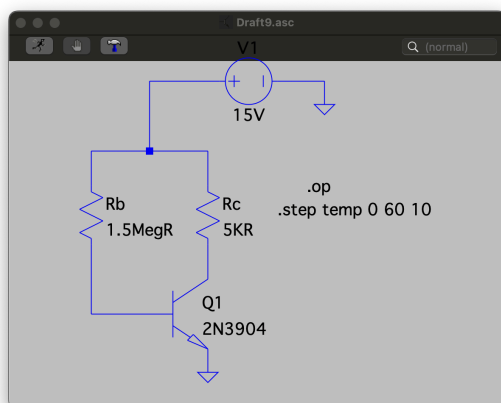
Experiment 1: Poor Biasing

Circuit Analysis:



Simulation:

a)



$$I_B = 9.54e-06$$

$$I_C = 2.85e-03$$

$$V_{CE} = 7.71e-01$$

$$V_{BE} = 6.83e-01$$

```
LTspice 17.2.4 for MacOS
Draft9.log
Circuit: *
Start time: Wed Jul 26 14:43:04 2024
Warning: Multiple definitions of model "2scr375p" Type: BJT
Warning: Multiple definitions of model "bc847b" Type: BJT
Warning: Multiple definitions of model "bc847c" Type: BJT
Warning: Multiple definitions of model "bc847d" Type: BJT
Solve: Normal
Maximum thread count: 8
temp = 27
method = modified trap
CompressMinPoints = 1824
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
--- Bipolar Transistors ---
Name: q1
Model: 2n3904
Ib: 9.54e-06
Ic: 2.85e-03
Vbe: 6.83e-01
Vbc: -9.85e-02
Vce: 7.71e-01
BetaDC: 2.98e+02
Gm: 1.99e-01
Rpi: 2.71e+03
Rr: 2.08e+01
Ro: 3.82e+04
Cbe: 5.18e-11
Cbc: 3.86e-12
Cjs: 0.00e+00
BetaAC: 2.98e+02
Cox: 0.00e+00
Ft: 3.17e+08

Operating Bias Point Solution:
V(n82) 9.77129 voltage
V(n83) 0.682806 voltage
V(n84) 15 voltage
Ic(Q1) 0.00284674 device_current
Ib(Q1) 9.5448e-06 device_current
Ic(Q1) -0.00285529 device_current
I(Rc) 0.00284674 device_current
I(Rb) 9.5448e-06 device_current
I(Q1) -0.00285529 device_current

Total elapsed time: 0.012 seconds.
```

b)

```

LTspice 17.2.4 for MacOS
Draft9.log
Circuit: * /Users/diegoceja/Documents/LTspice/Draft9.asc
Start Time: Wed Jul 24 19:45:44 2024
Warning: Multiple definitions of model "2scr375p" Type: BJT
Warning: Multiple definitions of model "bc857b" Type: BJT
Warning: Multiple definitions of model "bc847c" Type: BJT
Warning: Multiple definitions of model "bc847b" Type: BJT
solver = Normal
Maximum thread count: 8
tnom = 27
temp = 0
method = modified trap
CompressWinPoints = 1024
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
--- Bipolar Transistors ---
Name: q1
Model: 2n3904
Ib: 9.52e-06
Ic: 2.51e-03
Vbe: 7.25e-01
Vbc: -1.74e+00
Vce: 2.47e+00
BetaDC: 2.63e+02
Gm: 1.05e+01
Rpi: 2.47e+03
Rz: 2.80e+02
Ro: 4.80e+04
Cbe: 4.95e-11
Cbc: 2.64e-12
Cjs: 0.00e+00
BetaAC: 2.62e+02
Cvt: 0.00e+00
Ft: 3.23e+08

Operating Bias Point Solution:
V(n803) 2.46787 voltage
V(n804) 0.724592 voltage
V(n805) 15 voltage
V(n802) 15 voltage
Ic(Q1) 0.00258643 device_current
Ib(Q1) 9.517e-06 device_current
Ie(Q1) -0.00251594 device_current
I(Rc) 0.00258643 device_current
I(Rb) 9.517e-06 device_current
I(V1) -0.00258643 device_current
I(V2) -9.517e-06 device_current

Total elapsed time: 0.013 seconds.

```

```

LTspice 17.2.4 for MacOS
Draft9.log
Circuit: * /Users/diegoceja/Documents/LTspice/Draft9.asc
Start Time: Wed Jul 24 19:42:07 2024
Warning: Multiple definitions of model "2scr375p" Type: BJT
Warning: Multiple definitions of model "bc857b" Type: BJT
Warning: Multiple definitions of model "bc847c" Type: BJT
Warning: Multiple definitions of model "bc847b" Type: BJT
solver = Normal
Maximum thread count: 8
tnom = 27
temp = 60
method = modified trap
CompressWinPoints = 1024
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
--- Bipolar Transistors ---
Name: q1
Model: 2n3904
Ib: 9.58e-06
Ic: 2.96e-03
Vbe: 6.28e-01
Vbc: 6.43e-01
Vce: 1.86e-01
BetaDC: 3.07e+02
Gm: 1.02e+01
Rpi: 3.35e+03
Rz: 2.80e+02
Ro: 5.10e+03
Cbe: 4.91e-11
Cbc: 4.71e-11
Cjs: 0.00e+00
BetaAC: 3.43e+02
Cvt: 0.00e+00
Ft: 1.69e+08

Operating Bias Point Solution:
V(n803) 0.155677 voltage
V(n804) 0.62819 voltage
V(n805) 15 voltage
V(n802) 15 voltage
Ic(Q1) 0.00296286 device_current
Ib(Q1) 9.5812e-06 device_current
Ie(Q1) -0.00297245 device_current
I(Rc) 0.00296286 device_current
I(Rb) 9.5812e-06 device_current
I(V1) -0.00296286 device_current
I(V2) -9.5812e-06 device_current

Total elapsed time: 0.014 seconds.

```

Values	0° C	Default Temp (27° C)	60° C
I _B	9.52e-06	9.54e-06	9.52e-06
I _C	2.51e-03	2.85e-03	2.96e-03
V _{CE}	2.47e+00	7.71e-01	1.86e-01
V _{BE}	7.25e-01	6.83e-01	6.28e-01

Both currents I_B and I_C stay mostly the same, the only values that have a drastic change is V_{CE} and a more subtle change V_{BE} .

Experiment 2: Poor Biasing

Circuit Analysis:

a)

Assum BJT is in Active Mode

$$I_C = 150 I_B$$

$$V_{CE} \geq V_{D0}$$

Voltage Divider

$$V_B = 15 \frac{12K\Omega}{39K\Omega + 12K\Omega} \Rightarrow 15V \left(\frac{12}{51} \right) = 3.53V$$

$$V_E = V_B - V_{BE} \Rightarrow 3.53 - 0.7 = 2.83V$$

$$I_E = \frac{V_E}{R_E} = \frac{2.83V}{1K} = 2.83mA \quad \text{So } I_C = I_E$$

$$I_B = \frac{I_C}{\beta} = \frac{2.83mA}{150} = 18.87\mu A$$

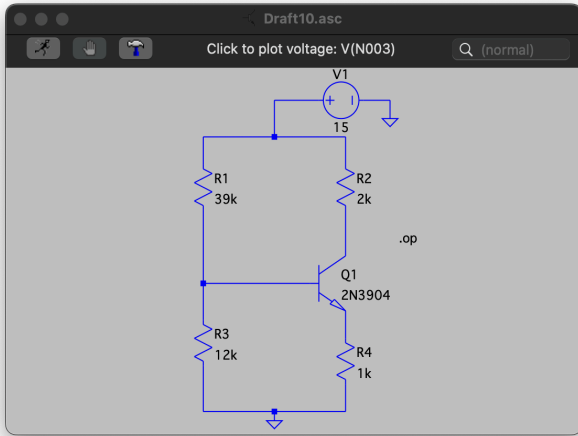
$$V_{RC} = I_C R_C = (2.83mA)(2K) = 5.66V \Rightarrow V_{CE} = V_{CC} - V_{RC} - V_E$$

$$= 15 - 5.66 - 2.83$$

$$V_{CE} = 6.51V$$

Simulation:

a)



$$I_B = 8.73e-06$$

$$I_C = 2.76e-03$$

$$V_{CE} = 6.71e+00$$

$$V_{BE} = 6.81e-01$$

```
Draft10.log

LTspice 17.2.4 for MacOS
Circuit: * /Users/diegoceja/Documents/LTspice/Draft10.asc
Start Time: Wed Jul 24 20:41:19 2024
Warning: Multiple definitions of model "2scr375p" Type: BJT
Warning: Multiple definitions of model "bc857b" Type: BJT
Warning: Multiple definitions of model "bc847c" Type: BJT
Warning: Multiple definitions of model "bc847b" Type: BJT
solver = Normal
Maximum thread count: 8
tnom = 27
temp = 27
method = modified trap
CompressWinPoints = 1024
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
--- Bipolar Transistors ---

Name: q1
Model: 2n3904
Ib: 8.73e-06
Ic: 2.76e-03
Vbe: 6.81e-01
Vbc: -6.83e+00
Vce: 6.71e+00
BetaDC: 3.16e+02
Gm: 1.00e-01
Rpi: 2.96e+03
Rx: 2.00e+01
Ro: 3.84e+04
Cbe: 4.99e-11
Cbc: 1.93e-12
Cjt: 0.00e+00
BetaAC: 3.16e+02
Cbx: 0.00e+00
Ft: 3.26e+08

Operating Bias Point Solution:
V(n001) 15 voltage
V(n003) 3.44927 voltage
V(n002) 9.47997 voltage
V(n004) 2.76375 voltage
Ic(Q1) 0.00276002 device_current
Ib(Q1) 8.73307e-06 device_current
Ie(Q1) -0.00276876 device_current
I(R1) 0.000295172 device_current
I(R2) 0.00276002 device_current
I(R3) 0.00267439 device_current
I(R4) 0.00276375 device_current
I(V1) -0.00305619 device_current

Total elapsed time: 0.012 seconds.
```

b)

```
Draft10.log

LTspice 17.2.4 for MacOS
Circuit: * /Users/diegoceja/Documents/LTspice/Draft10.asc
Start Time: Wed Jul 24 20:41:19 2024
Warning: Multiple definitions of model "2scr375p" Type: BJT
Warning: Multiple definitions of model "bc857b" Type: BJT
Warning: Multiple definitions of model "bc847c" Type: BJT
Warning: Multiple definitions of model "bc847b" Type: BJT
solver = Normal
Maximum thread count: 8
tnom = 27
temp = 0
method = modified trap
CompressWinPoints = 1024
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
--- Bipolar Transistors ---

Name: q1
Model: 2n3904
Ib: 9.84e-06
Ic: 2.70e-03
Vbe: 7.25e-01
Vbc: -6.15e+00
Vce: 6.88e+00
BetaDC: 2.75e+02
Gm: 1.14e-01
Rpi: 2.39e+03
Rx: 2.00e+01
Ro: 3.92e+04
Cbe: 5.25e-11
Cbc: 1.91e-12
Cjt: 0.00e+00
BetaAC: 2.73e+02
Cbx: 0.00e+00
Ft: 3.34e+08

Operating Bias Point Solution:
V(n001) 15 voltage
V(n003) 3.43999 voltage
V(n002) 9.59219 voltage
V(n004) 2.71375 voltage
Ic(Q1) 0.00270391 device_current
Ib(Q1) 9.84225e-06 device_current
Ie(Q1) -0.00271375 device_current
I(R1) 0.000294433 device_current
I(R2) 0.00270391 device_current
I(R3) 0.00265591 device_current
I(R4) 0.00271375 device_current
I(V1) -0.00300034 device_current

Total elapsed time: 0.011 seconds.
```

```
Draft10.log

LTspice 17.2.4 for MacOS
Circuit: * /Users/diegoceja/Documents/LTspice/Draft10.asc
Start Time: Wed Jul 24 20:41:49 2024
Warning: Multiple definitions of model "2scr375p" Type: BJT
Warning: Multiple definitions of model "bc857b" Type: BJT
Warning: Multiple definitions of model "bc847c" Type: BJT
Warning: Multiple definitions of model "bc847b" Type: BJT
solver = Normal
Maximum thread count: 8
tnom = 27
temp = 60
method = modified trap
CompressWinPoints = 1024
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
--- Bipolar Transistors ---

Name: q1
Model: 2n3904
Ib: 7.46e-06
Ic: 2.83e-03
Vbe: 6.25e-01
Vbc: -5.89e+00
Vce: 6.51e+00
BetaDC: 3.69e+02
Gm: 9.68e-02
Rpi: 3.74e+03
Rx: 2.00e+01
Ro: 3.74e+04
Cbe: 4.74e-11
Cbc: 1.96e-12
Cjt: 0.00e+00
BetaAC: 3.66e+02
Cbx: 0.00e+00
Ft: 3.16e+08

Operating Bias Point Solution:
V(n001) 15 voltage
V(n003) 3.45912 voltage
V(n002) 9.34788 voltage
V(n004) 2.83412 voltage
Ic(Q1) 0.00282444 device_current
Ib(Q1) 7.65946e-06 device_current
Ie(Q1) -0.00283412 device_current
I(R1) 0.00029592 device_current
I(R2) 0.00282444 device_current
I(R3) 0.00282626 device_current
I(R4) 0.00283412 device_current
I(V1) -0.00312238 device_current

Total elapsed time: 0.011 seconds.
```

Values	0° C	25° C	Default Temp (27° C)	60° C
I_B	9.58e-06	8.81e-06	8.73e-06	7.66e-06
I_C	2.70e-03	2.76e-03	2.76e-03	2.83e-03
V_{CE}	6.88e+00	6.72e+00	6.71e+00	6.51e+00
V_{BE}	7.25e-01	6.84e-01	6.81e-01	6.25e-01

```

LTspice 17.2.4 for MacOS
Circuit: * /Users/giangocja/Documents/LTspice/Draft10.asc
Start Time: Wed Jul 24 20:41:31 2024
Warning: Multiple definitions of model "2scr375p" Type: BJT
Warning: Multiple definitions of model "bc857b" Type: BJT
Warning: Multiple definitions of model "bc847c" Type: BJT
Warning: Multiple definitions of model "bc847b" Type: BJT
Solver = Normal
Maximum thread count: 8
tnom = 27
temp = 25
method = modified trap
CompressWinPoints = 1024
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
--- Bipolar Transistors ---

Name: q1
Model: 2n3904
Ib: 8.81e-06
Ic: 2.76e-03
Vbc: 6.84e-01
Vbc: -6.84e+00
Vce: 6.72e+00
BetaDC: 3.11e+02
Gm: 1.07e-01
Rpi: 2.91e+03
Rr: 2.00e+01
Ro: 3.04e+04
Cbe: 5.01e-11
Cbci: 1.93e-12
Cjs: 0.00e+00
BetaAC: 3.11e+02
Cbv: 0.00e+00
Ft: 3.26e+08

Operating Bias Point Solution:
V(n001) 15 voltage
V(n003) 3.44859 voltage
V(n002) 9.48816 voltage
V(n004) 2.76673 voltage
Ic(Q1) 0.0027592 device_current
Ib(Q1) 8.80719e-06 device_current
Ie(Q1) -0.00276473 device_current
I(R1) 0.00029619 device_current
I(R2) 0.0027592 device_current
I(R3) 0.000287383 device_current
I(R4) 0.00276473 device_current
I(V1) -0.00305211 device_current

Total elapsed time: 0.011 seconds.

```

Current I_B seems to have a gradual decrease as it increases in temp going from 9.58e-06 at 0C to 7.66e-06 at 60C. Current I_C has a slight increase in current as it increases in temperature. Both V_{CE} and V_{BE} have a slight decrease in voltage as temperature increases. Compared to experiment 1, the biggest difference would be that both currents I_B and I_C stay mostly the same in experiment 1 whereas in experiment 2 both have changes in their current.